

ARID DESERT LANDFORMS / THAR DESERTIFICATION

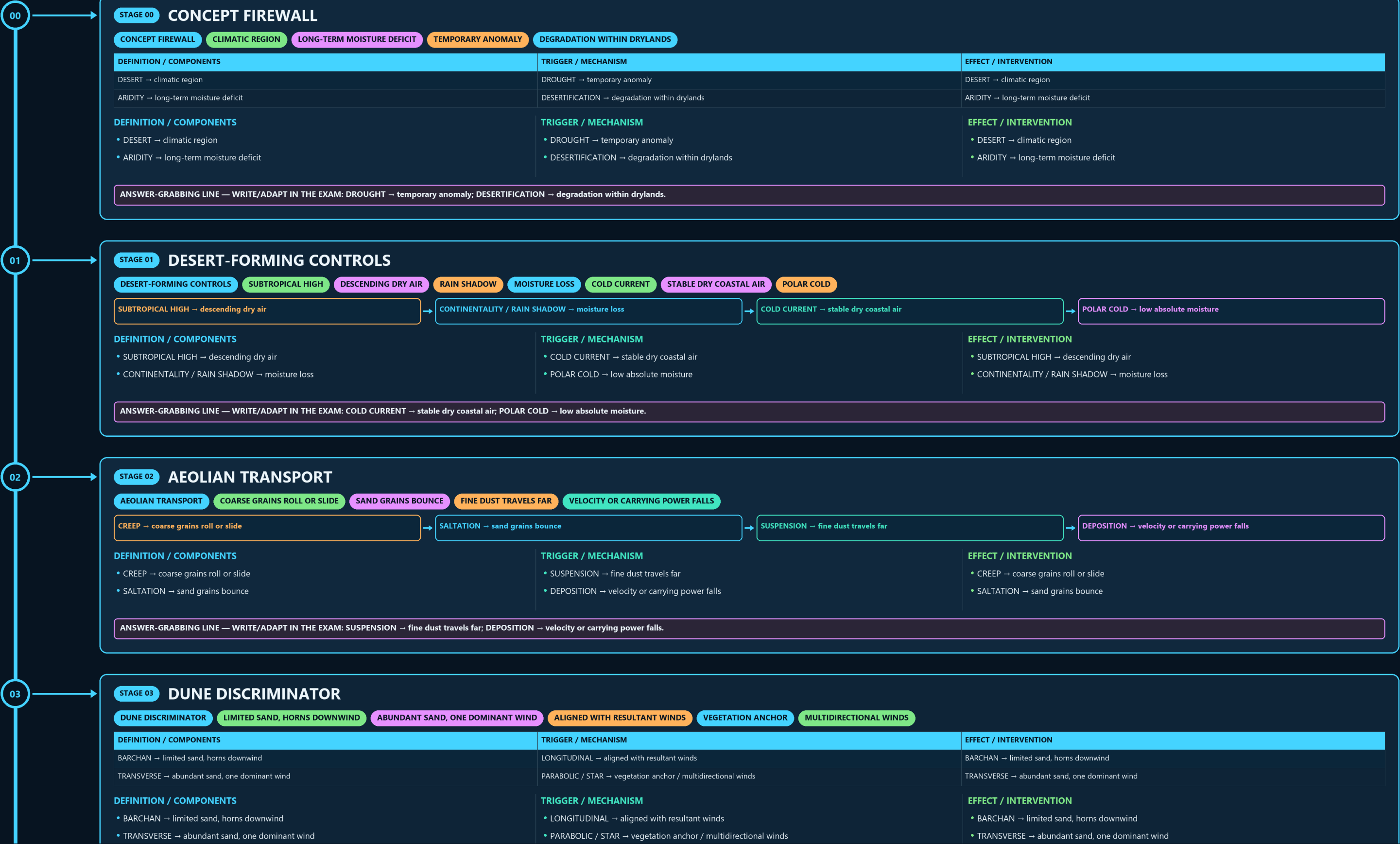
CONCEPT FIREWALL → DESERT-FORMING CONTROLS → AEOLIAN TRANSPORT → DUNE DISCRIMINATOR → WATER IN THE DESERT → THAR SPATIAL ANATOMY

Read the thick cyan rail from Stage 00 to the final core synthesis. Each card uses a concept-appropriate structure; the grey E card is optional enrichment only and never repairs missing core content.

PRIMARY CORE FLOW SUBORDINATE EXTRA APPROVAL / REVIEW

Approval: FALSE • Pending user review • active generation g2 • explicit approval of this exact generation is required • all prior artifacts unchanged

CORE BEFORE EXTRA • prior artefacts unchanged • exact same master drives poster and tiles



03

SALTATION → sand grains bounce

DEPOSITION → velocity or carrying power falls

SALTATION → sand grains bounce

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: SUSPENSION → fine dust travels far; DEPOSITION → velocity or carrying power falls.

04

STAGE 03

DUNE DISCRIMINATOR

DUNE DISCRIMINATOR

LIMITED SAND, HORNS DOWNWIND

ABUNDANT SAND, ONE DOMINANT WIND

ALIGNED WITH RESULTANT WINDS

VEGETATION ANCHOR

MULTIDIRECTIONAL WINDS

DEFINITION / COMPONENTS	TRIGGER / MECHANISM	EFFECT / INTERVENTION
BARCHAN → limited sand, horns downwind	LONGITUDINAL → aligned with resultant winds	BARCHAN → limited sand, horns downwind
TRANSVERSE → abundant sand, one dominant wind	PARABOLIC / STAR → vegetation anchor / multidirectional winds	TRANSVERSE → abundant sand, one dominant wind

DEFINITION / COMPONENTS

• BARCHAN → limited sand, horns downwind

• TRANSVERSE → abundant sand, one dominant wind

TRIGGER / MECHANISM

• LONGITUDINAL → aligned with resultant winds

• PARABOLIC / STAR → vegetation anchor / multidirectional winds

EFFECT / INTERVENTION

• BARCHAN → limited sand, horns downwind

• TRANSVERSE → abundant sand, one dominant wind

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: PARABOLIC / STAR → vegetation anchor / multidirectional winds.

05

STAGE 04

WATER IN THE DESERT

WATER IN THE DESERT

STORM RUNOFF

WADI DISCHARGE

MOUNTAIN FRONT

ALLUVIAL FAN

FANS COALESCE

CLOSED BASIN

PLAYA AND SALT CONCENTRATION

STORM RUNOFF → wadi discharge

MOUNTAIN FRONT → alluvial fan

FANS COALESCE → bajada

CLOSED BASIN → playa and salt concentration

DEFINITION / COMPONENTS	TRIGGER / MECHANISM	EFFECT / INTERVENTION
• STORM RUNOFF → wadi discharge	• FANS COALESCE → bajada	• STORM RUNOFF → wadi discharge
• MOUNTAIN FRONT → alluvial fan	• CLOSED BASIN → playa and salt concentration	• MOUNTAIN FRONT → alluvial fan

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: FANS COALESCE → bajada; CLOSED BASIN → playa and salt concentration.

06

STAGE 05

THAR SPATIAL ANATOMY

THAR SPATIAL ANATOMY

SANDY MARUSTHALI CORE

BAGAR AND ARAVALLI TRANSITION

LUNI AND SALINE-BASIN ROUTE

CANAL-ALLUVIAL TRANSFORMATION

EAST → Bagar and Aravalli transition

WEST → sandy Marusthali core

SOUTHWEST → Luni and saline-basin route

NORTH → canal-alluvial transformation

SYSTEM / DEFINITION	PROCESS / MECHANISM	SPATIAL / INDIA PATTERN	HAZARD / LIMIT / RESPONSE
• WEST → sandy Marusthali core	• SOUTHWEST → Luni and saline-basin route	• WEST → sandy Marusthali core	• SOUTHWEST → Luni and saline-basin route
• EAST → Bagar and Aravalli transition	• NORTH → canal-alluvial transformation	• EAST → Bagar and Aravalli transition	• NORTH → canal-alluvial transformation

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: SOUTHWEST → Luni and saline-basin route; NORTH → canal-alluvial transformation.

STAGE 06

SUSCEPTIBILITY-TO-DEGRADATION

VARIABLE RAIN + WIND

NATURAL STRESS

COVER LOSS + SOIL DISTURBANCE

WATER MISMANAGEMENT

SALINITY OR DEPLETION

THRESHOLD CROSSED

RECOVERY CAPACITY DECLINES

VARIABLE RAIN + WIND → natural stress

COVER LOSS + SOIL DISTURBANCE → exposure

WATER MISMANAGEMENT → salinity or depletion

THRESHOLD CROSSED → recovery capacity declines

DEFINITION / COMPONENTS	TRIGGER / MECHANISM	EFFECT / INTERVENTION
• VARIABLE RAIN + WIND → natural stress	• WATER MISMANAGEMENT → salinity or depletion	• VARIABLE RAIN + WIND → natural stress
• COVER LOSS + SOIL DISTURBANCE → exposure	• THRESHOLD CROSSED → recovery capacity declines	• COVER LOSS + SOIL DISTURBANCE → exposure

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: WATER MISMANAGEMENT → salinity or depletion; THRESHOLD CROSSED → recovery capacity declines.

ARID DESERT LANDFORMS / THAR DESERTIFICATION • TILE 2/4 • same master rows 2423-5657 of 10502 • continues below

06

STAGE 06 SUSCEPTIBILITY-TO-DEGRADATION

VARIABLE RAIN + WIND NATURAL STRESS COVER LOSS + SOIL DISTURBANCE WATER MISMANAGEMENT SALINITY OR DEPLETION THRESHOLD CROSSED RECOVERY CAPACITY DECLINES

VARIABLE RAIN + WIND → natural stress

COVER LOSS + SOIL DISTURBANCE → exposure

WATER MISMANAGEMENT → salinity or depletion

THRESHOLD CROSSED → recovery capacity declines

DEFINITION / COMPONENTS

- VARIABLE RAIN + WIND → natural stress
- COVER LOSS + SOIL DISTURBANCE → exposure

TRIGGER / MECHANISM

- WATER MISMANAGEMENT → salinity or depletion
- THRESHOLD CROSSED → recovery capacity declines

EFFECT / INTERVENTION

- VARIABLE RAIN + WIND → natural stress
- COVER LOSS + SOIL DISTURBANCE → exposure

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: WATER MISMANAGEMENT → salinity or depletion; THRESHOLD CROSSED → recovery capacity declines.

07

STAGE 07 ARAVALLI BOUNDARY

ARAVALLI BOUNDARY MODIFIES LOCAL WIND AND RUNOFF DIVIDES AND ROUTES WATER SUPPORTS CONNECTIVITY NOT AN IMPERMEABLE DESERT WALL

DEFINITION / COMPONENTS

RELIEF → modifies local wind and runoff

DRAINAGE → divides and routes water

TRIGGER / MECHANISM

HABITAT → supports connectivity

LIMIT → not an impermeable desert wall

EFFECT / INTERVENTION

RELIEF → modifies local wind and runoff

DRAINAGE → divides and routes water

DEFINITION / COMPONENTS

- RELIEF → modifies local wind and runoff
- DRAINAGE → divides and routes water

TRIGGER / MECHANISM

- HABITAT → supports connectivity
- LIMIT → not an impermeable desert wall

EFFECT / INTERVENTION

- RELIEF → modifies local wind and runoff
- DRAINAGE → divides and routes water

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: HABITAT → supports connectivity; LIMIT → not an impermeable desert wall.

08

STAGE 08 CANAL BALANCE

CANAL BALANCE WATER SUPPLY AND PRODUCTION WATER-TABLE RISE POOR DRAINAGE WATERLOGGING AND SALINITY CROP-WATER FIT + DRAINAGE + MONITORING

DEFINITION / COMPONENTS

BENEFIT → water supply and production

SEEPAGE → water-table rise

TRIGGER / MECHANISM

POOR DRAINAGE → waterlogging and salinity

MANAGEMENT → crop-water fit + drainage + monitoring

EFFECT / INTERVENTION

BENEFIT → water supply and production

SEEPAGE → water-table rise

DEFINITION / COMPONENTS

- BENEFIT → water supply and production
- SEEPAGE → water-table rise

TRIGGER / MECHANISM

- POOR DRAINAGE → waterlogging and salinity
- MANAGEMENT → crop-water fit + drainage + monitoring

EFFECT / INTERVENTION

- BENEFIT → water supply and production
- SEEPAGE → water-table rise

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: MANAGEMENT → crop-water fit + drainage + monitoring.

09

STAGE 09 LDN RESPONSE HIERARCHY

LDN RESPONSE HIERARCHY PROTECT FUNCTIONING LAND IMPROVE GRAZING, TILLAGE AND WATER USE RESTORE SOIL, GRASS AND DRAINAGE MONITOR GAINS AND LOSSES TRANSPARENTLY

AVOID → protect functioning land

REDUCE → improve grazing, tillage and water use

REVERSE → restore soil, grass and drainage

BALANCE → monitor gains and losses transparently

DEFINITION / COMPONENTS

- AVOID → protect functioning land
- REDUCE → improve grazing, tillage and water use

TRIGGER / MECHANISM

- REVERSE → restore soil, grass and drainage
- BALANCE → monitor gains and losses transparently

EFFECT / INTERVENTION

- AVOID → protect functioning land
- REDUCE → improve grazing, tillage and water use

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: REDUCE → improve grazing, tillage and water use.

09

STAGE 09

LDN RESPONSE HIERARCHY

LDN RESPONSE HIERARCHY

PROTECT FUNCTIONING LAND

IMPROVE GRAZING, TILLAGE AND WATER USE

RESTORE SOIL, GRASS AND DRAINAGE

MONITOR GAINS AND LOSSES TRANSPARENTLY

AVOID → protect functioning land

REDUCE → improve grazing, tillage and water use

REVERSE → restore soil, grass and drainage

BALANCE → monitor gains and losses transparently

DEFINITION / COMPONENTS

• AVOID → protect functioning land

• REDUCE → improve grazing, tillage and water use

TRIGGER / MECHANISM

• REVERSE → restore soil, grass and drainage

• BALANCE → monitor gains and losses transparently

EFFECT / INTERVENTION

• AVOID → protect functioning land

• REDUCE → improve grazing, tillage and water use

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: REDUCE → improve grazing, tillage and water use.

10

STAGE 10

RESTORATION MATRIX

RESTORATION MATRIX

WIND EROSION

COVER, SHELTER AND DUNE MANAGEMENT

RUNOFF EROSION

CONTOUR AND WATERSHED TREATMENT

DRAINAGE AND WATER BALANCE

RANGELAND DECLINE

NATIVE GRASS + INSTITUTIONS

CLASSIFICATION	DIAGNOSTIC BASIS	PROCESS LINK	EXAM TRAP
WIND EROSION → cover, shelter and dune management	SALINITY → drainage and water balance	WIND EROSION → cover, shelter and dune management	SALINITY → drainage and water balance
RUNOFF EROSION → contour and watershed treatment	RANGELAND DECLINE → native grass + institutions	RUNOFF EROSION → contour and watershed treatment	RANGELAND DECLINE → native grass + institutions

CLASSIFICATION

• WIND EROSION → cover, shelter and dune management

• RUNOFF EROSION → contour and watershed treatment

DIAGNOSTIC BASIS

• SALINITY → drainage and water balance

• RANGELAND DECLINE → native grass + institutions

PROCESS LINK

• WIND EROSION → cover, shelter and dune management

• RUNOFF EROSION → contour and watershed treatment

EXAM TRAP

• SALINITY → drainage and water balance

• RANGELAND DECLINE → native grass + institutions

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: WIND EROSION → cover, shelter and dune management.

11

STAGE 11

DESERTIFICATION ANSWER SPINE

DESERTIFICATION ANSWER SPINE

RETAIN UNCCD DRYLAND BOUNDARY

LOCATE PROCESS AND PRESSURE

SUSCEPTIBILITY PLUS ACCELERATOR

PROCESS-MATCHED, MONITORED AND QUALIFIED

DEFINE → retain UNCCD dryland boundary

MAP → locate process and pressure

DIAGNOSE → susceptibility plus accelerator

RESPOND → process-matched, monitored and qualified

DEFINITION / COMPONENTS

• DEFINE → retain UNCCD dryland boundary

• MAP → locate process and pressure

TRIGGER / MECHANISM

• DIAGNOSE → susceptibility plus accelerator

• RESPOND → process-matched, monitored and qualified

EFFECT / INTERVENTION

• DEFINE → retain UNCCD dryland boundary

• MAP → locate process and pressure

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: DIAGNOSE → susceptibility plus accelerator; RESPOND → process-matched, monitored and qualified.

E

EXTRA

SUBORDINATE ENRICHMENT — ONLY AFTER THE COMPLETE CORE

OPTIONAL ADVANCED ENRICHMENT

THE THAR LIES WEST OF THE ARAVALLI, BETWEEN THE

BOOK ESTIMATES OF DESERT AREA VARY WITH THE BOUNDARY

THE RANN OF KACHCHH IS TREATED AS AN EXTENSION

ECONOMY (MAJID)

BAJRA, PULSES, FODDER; LIVESTOCK IS CENTRAL; SCOPE FOR DATE-PALM

IT IS DESCRIBED AS THE WESTERN DRY REGION —

~29.7% OF INDIA (~97.85 MHA) IS DEGRADED (2018-19), UP

OPTIONAL PROCESS DEPTH

• The Thar lies west of the Aravalli, between the Indus and the Aravalli, spilling into Punjab, Haryana and northern Gujarat.

• Book estimates of desert area vary with the boundary used; retain them as source-era figures, not

• The Rann of Kachchh is treated as an extension of the desert; Jaisalmer and Barmer show the strongest aeolian (dune) topography.

• Economy (Majid): bajra, pulses, fodder; livestock is central; scope for date-palm, water-melon and rain-water harvesting.

DATA / INSTITUTION LIMIT

• It is described as the Western Dry Region — an arid tract (Marwar/Mewar margins) with extreme temperature range.

• ~29.7% of India (~97.85 mha) is degraded (2018-19), up from 28.76% (2003-05) — a rising trend.

• Rajasthan has the largest desertified area; wind erosion drives 40%+ of its desertification.

• India targets Land Degradation Neutrality (LDN) by 2030 under the UNCCD; tools include Green India Mission and watershed/afforestation programmes.

CONTEMPORARY PARALLEL

• Thar = west of Aravalli, between Indus & Aravalli.

• Area figures vary by delimitation; western core is arid and margins semi-arid.

• Rann of Kachchh is linked to the wider arid region but is a distinct saline mudflat/marsh.

• Jaisalmer & Barmer = strongest aeolian (dune) topography.

OPTIONAL STATUS — This optional depth is unnecessary for a competent core answer; use it only for added nuance after the complete core has been written.